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LIFT ARM ATTACHMENT SAFETY SYSTEM

Abstract

A lift arm attachment safety system may include, an adapter plate receivable within an adapter frame of a lift arm attachment; one or more configured to detect a proximity of the adapter plate to the adapter frame; a drive control system configured to control drive capabilities of a vehicle; at least one processor; and at least one memory device storing one or more instructions for: receiving at least one signal from the one or more sensors indicating that the adapter plate is within a threshold proximity to the adapter frame; transmitting one or more control signals to the drive control system to cause the drive control system to disable drive capabilities of the vehicle in response to receiving the at least one signal from the one or more sensors indicating that the adapter plate is within a threshold proximity to the adapter frame.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure generally relates to the field of lift-arm vehicle safety systems, and more particularly, to a system of sensors and vehicle control systems configured to ensure safe operation of the lift-arm vehicle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] The numerous advantages of the disclosure may be better understood by those skilled in the art by reference to the accompanying figures in which:

[0003] FIG. **1** shows a schematic view of a lift arm attachment safety system;

[0004] FIG. **2** shows an environmental view of a lift arm attachment safety system;

[0005] FIG. **3** shows a lift arm attachment;

[0006] FIG. **4** shows an adapter plate including one or more sensors of a lift arm attachment safety system;

[0007] FIG. **5** shows an adapter plate including one or more sensors of a lift arm attachment safety system;

[0008] FIG. **6** shows a mechanism for coupling an adapter plate to a lift arm attachment;

[0009] FIG. **7** shows a mechanism for coupling an adapter plate to a lift arm attachment;

[0010] FIG. **8** shows a mechanism for coupling an adapter plate to a lift arm attachment;

[0011] FIG. **9** shows a mechanism for coupling an adapter plate to a lift arm attachment;

[0012] FIG. **10** shows a mechanism for coupling an adapter plate to a lift arm attachment;

[0013] FIG. **11** shows a mechanism for coupling an adapter plate to a lift arm attachment;

[0014] FIG. **12** shows a mechanism for coupling an adapter plate to a lift arm attachment;

[0015] FIG. **13** shows a logic diagram of a lift arm attachment safety system;

[0016] FIG. **14** shows a logic diagram of a lift arm attachment safety system;

[0017] FIG. **15** shows a logic diagram of a lift arm attachment safety system;

[0018] FIG. **16** shows a logic diagram of a lift arm attachment safety system;

[0019] FIG. **17** shows a logic diagram of a lift arm attachment safety system; and

[0020] FIG. **18** shows a logic diagram of a lift arm attachment safety system;

DETAILED DESCRIPTION

[0021] The present disclosure has been particularly shown and described with respect to certain embodiments and specific features thereof. The embodiments set forth herein are taken to be illustrative rather than limiting. It should be readily apparent to those of ordinary skill in the art that various changes and modifications in form and detail may be made without departing from the spirit and scope of the disclosure.

[0022] Reference will now be made in detail to the subject matter disclosed, which is illustrated in the accompanying drawings. Referring generally to FIGS. **1** through **18**, embodiments of the present disclosure are generally directed to a lift arm attachment safety system **100**.

[0023] Referring to FIG. **1**, a schematic view of the control components of the lift arm attachment safety system **100** is shown. The lift arm attachment safety system **100** may include one or more sensors **101** (e.g., first sensors **101A**, second sensors **101B** and third sensors **101C**) a processor **102** (e.g., programmable logic computing devices including memory configured for storing computer readable instructions, application-specific microcontrollers, relays, and the like). Based on one or more outputs of the sensors **101**, the lift arm attachment safety system **100** may provide control signals to locking mechanism control circuitry **103** configured to enable or disable a locking

mechanism **104** to secure a lift arm attachment **110** to a vehicle **105** (e.g., as shown in FIG. 2). Further, based on one or more outputs of the sensors **101**, the lift arm attachment safety system **100** may provide control signals to a vehicle drive control circuitry **106** configured to enable control of a drive control system **107** (e.g., a parking break, a transmission disengagement, a hydraulic system, or any other system configured to disable drive functionality of the vehicle **105** as described below and in FIGS. **13-18**), of the vehicle **105**.

[0024] While described herein in the context of a compact skid-steer loader or track loader-type vehicle **105** as shown in FIG. 2 (e.g., vehicles such as those manufactured and sold by John Deere®, Bobcat®, Kubota®, Case®, Caterpillar®, Ditch Witch®, New Holland®, and the like), the technologies disclosed herein will be recognized as being extensible to any lift-arm equipped vehicle (e.g., bucket loaders, backhoes, tractor front-end loaders, combines, excavators, dozers, fork lifts, cranes, mowers, and the like) which are configured to receive and/or couple to various attachments, accessories, or implements (loader buckets, excavator buckets, snowplows, mower heads, combine heads, and the like).

[0025] The lift arm attachment safety system **100** may further include an interface device **124** (e.g., an LED indicator, one or more physical buttons, touch screen, gesture recognition, voice recognition, and the like) configured to allow an operator to interact with the lift arm attachment safety system **100**. The interface device **124** may receive signals from the processor **102** and display status information for the lift arm attachment safety system **100** to the operator. Further, the interface device **124** may receive user inputs from the operator to control various operations of the lift arm attachment safety system **100**.

[0026] Referring to FIG. 2, a lift arm attachment safety system **100** may include an adapter plate **108** which may be coupled to the lift arms **109** of the vehicle **105**. The adapter plate **108** may be manipulated by an operator via the lift arms **109** to engage with and connect to a lift arm attachment **110** (e.g., a loader bucket).

[0027] Referring to FIG. 3, in one example, the lift arm attachment **110** may include an adapter frame **111** including back plate **112** (e.g., a rear surface of the lift arm attachment **110** or a separate panel affixed to the rear surface of the lift arm attachment **110**), a top flange portion **113** and a bottom flange portion **114**. The bottom flange portion **114** may include one or more locking pin apertures **115**.

[0028] Referring to FIGS. 4-6, the lift arm attachment safety system **100** may include the adapter plate **108** configured to engage the top flange portion **113** of the adapter frame **111** of the lift arm attachment **110**. In one example, the adapter plate **108** may include one or more projection portions **116** having one or more surfaces **117** which are disposed at an angle relative to the adapter plate **108**. For example, the surfaces **117** of the projection portions **116** may be configured such that upon insertion of the projection portions **116** under the top flange portion **113** of the adapter frame **111** and drawing the adapter plate **108** into contact with the back plate **112** of the adapter frame **111**, the surfaces **117** are substantially parallel to the top flange portion **113**. Such a configuration is further described with respect to FIGS. 6-9 below.

[0029] Referring to FIGS. 3 and 4, The lift arm attachment safety system **100** may further include a locking mechanism **104**. The locking mechanism **104** may include one or more locking pin apertures **118** disposed in one or more flange portions **119** of the adapter plate **108**. The locking mechanism **104** may include one or more actuators (e.g., hydraulic actuators, electric linear actuators, and the like) (not shown) configured to extend or retract one or more locking pins **120** through the locking pin apertures **118** in the flange portions **119** to engage the locking pin apertures **115** of the bottom flange portion **114** of the adapter frame **111** of the lift arm attachment **110** to couple the adapter plate **108** to the lift arm attachment **110**. Such a configuration is further described with respect to FIGS. 6-9 below.

[0030] As shown in FIG. 1, the lift arm attachment safety system **100** may further include one or more sensors **101**. The sensors **101** may be contact-based or contactless positional

location/proximity sensors. The various examples, the sensors **101** may include, but are not limited to, inductive, eddy-current, capacitive, magnetorestrictive, piezoelectric, Hall effect, fiber-optic, optical, electromechanical, and/or ultrasonic-type sensors, and the like.

[0031] In one example, as shown in FIG. 4, a set of one or more first sensors **101A** may be disposed in one or more recesses **121** in the projection portions **116** of the adapter plate **108**. Such a configuration allows the first sensors **101A** to determine whether the projection portions **116** are within a threshold proximity (e.g., less than 1 inch) relative to the top flange portion **113** of the adapter frame **111** when the adapter plate **108** is inserted into the adapter frame **111** (as further shown in FIGS. 6-11), thereby allowing for the lift arm attachment safety system **100** to detect proper engagement of the adapter plate **108** with the adapter frame **111**. For example, the first sensors **101A** may be disposed on the adapter plate **108** and configured such that their sensor medium (e.g., light beam, electromagnetic field, mechanical switch) projects from the first sensors **101A** through the recesses **123** to determine a proximity of the adapter plate **108** to the top flange portion **113** of the adapter frame **111**.

[0032] Further, referring to FIG. 5, the adapter plate **108** may include one or more sensor apertures **122** through the adapter plate **108**. A set of one or more second sensors **101B** may be disposed on the adapter plate **108** and configured such that their sensor medium (e.g., light beam, electromagnetic field, mechanical switch) projects from the second sensors **101B** through the sensor apertures **122** to determine a proximity of the adapter plate **108** to the back plate **112** of the adapter frame **111** as the adapter plate **108** is inserted into the adapter frame **111** thereby allowing for the lift arm attachment safety system **100** to further detect proper engagement of the adapter plate **108** with the adapter frame **111**.

[0033] Still further, referring to FIG. 4, set of one or more third sensors **101C** may be disposed on the adapter plate **108** and configured such that their sensor medium (e.g., light beam, electromagnetic field, mechanical switch) projects from the third sensors **101C** across a path of movement of the locking pins **120** to detect a position of the locking pins **120**. In one example, the third sensors **101C** may detect if the locking pins **120** are in an extended or retracted position. In another example, the locking pin detection functionality of the third sensors **101C** may be implemented in the processor **102** of FIG. 1 or in a dedicated controller for actuators for the locking mechanism **104** where the controller maintains data regarding the status of the locking pins **120** (e.g., % extended or retracted) which may be queried by other components of the lift arm attachment safety system **100**.

[0034] Referring to FIGS. 6-9, operations related to engagement between the adapter plate **108** and the adapter frame **111** of the lift arm attachment **110** are shown. In FIG. 6, an operator may control the movement of the vehicle **105** or its lift arms **109** (as shown in FIG. 2) to move the adapter plate **108** forward to the adapter frame **111** such that the projection portions **116** pass under top flange portion **113**. In FIG. 7, the adapter plate **108** is moved slightly upward such that the projection portions **116** are positioned within one or more recesses **123** defined between the top flange portion **113** and the back plate **112** of the adapter frame **111**. In FIG. 8, the bottom end of the adapter plate **108** may be moved forward causing the projection portions **116** to rotate within the recesses **123** until, as shown in FIG. 9, the projection portions **116** abut the top flange portion **113** and the adapter plate **108** and the flange portions **119** abut the back plate **112** and the bottom flange portion **114**, respectively. In FIG. 9, an operator may control the movement of the vehicle **105** or its lift arms **109** to lift the lift arm attachment **110** off the ground to fully seat the adapter plate **108** within the adapter frame **111**. In FIGS. 9-12, following seating of the adapter plate **108** within the adapter frame **111**, the locking mechanism **104** is in the engaged position to cause the actuators to extend the locking pins **120** through both the locking pin apertures **118** of the flange portions **119** and the locking pin apertures **115** of the bottom flange portion **114** of the adapter frame **111** so as to secure the adapter plate **108** to the adapter frame **111** of the lift arm attachment **110**. In one example, the processor **102** may automatically transmit one or more control signals (e.g., following a delay

period of from 1 to 10 seconds) via the locking mechanism control circuitry **103** to the locking mechanism **104** (e.g., as shown in FIG. **1**), responsive to a detection of the seating of the adapter plate **108** within the adapter frame **111** by the second sensors **101B**, and instructing the locking mechanism **104** to extend the locking pins **120** into their locked position. Alternately, the processor **102** may receive one or more signals from the second sensors **101B** indicating a detection of the seating of the adapter plate **108** within the adapter frame **111**. The processor **102** may provide one or more signals to the interface device **124** causing the interface device **124** to display a notification to the operator that the adapter plate **108** within the adapter frame **111**. The operator may provide a user input via the interface device **124** requesting the engagement of the locking mechanism **104**. Responsive to the operator request, the processor **102** may transmit one or more control signals to the locking mechanism **104** instructing the locking mechanism **104** to extend the locking pins **120** into their locked position

[0035] FIGS. **13-19** depict logic diagrams representing the various control states of the lift arm attachment safety system **100** shown in FIG. **1** in controlling activation and deactivation of the locking mechanism **104** and the drive control system **107**.

[0036] In FIG. **13**, the lift arm attachment safety system **100** is shown in a state where the adapter plate **108** has not engaged the adapter frame **111** of the lift arm attachment **110** (e.g., as shown in FIG. **6**) such that the first sensors **101A** do not indicate that the projection portions **116** are in proximity to the top flange portion **113**.

[0037] In FIG. **14**, the lift arm attachment safety system **100** is shown in a state where the adapter plate **108** has engaged the adapter frame **111** of the lift arm attachment **110** (e.g., as shown in FIGS. **7** and **8**) such that the first sensors **101A** indicate that the projection portions **116** are in proximity to the top flange portion **113**. At this point, the lift arm attachment safety system **100** may transmit one or more control signals via the vehicle drive control circuitry **106** to the drive control system **107** instructing the drive control system **107** to, for example, engage a parking brake of the vehicle **105**. Preventing movement of the vehicle **105** until the lift arm attachment safety system **100** determines that the adapter plate **108** has properly engaged the adapter frame **111** of the lift arm attachment **110** and that the locking pins **120** are in place greatly reduces the risk of injury to operators of the vehicle **105** and/or other personnel in the area of the vehicle **105**.

[0038] In FIG. **15**, the lift arm attachment safety system **100** is shown in a state where the adapter plate **108** has been fully seated within the adapter frame **111** of the lift arm attachment **110** (e.g., as shown in FIG. **9**) such that the second sensors **101B** indicate that the adapter plate **108** (e.g., the flange portions **119**) is in proximity to the back plate **112** (e.g., the bottom flange portion **114**). At this point, either automatically or in response to a user request received via the interface device **124**, the processor **102** may transmit one or more control signals via the locking mechanism control circuitry **103** to the locking mechanism **104** (e.g., as shown in FIG. **1**) instructing the locking mechanism **104** to extend the locking pins **120** into their locked position.

[0039] In FIG. **16**, the lift arm attachment safety system **100** is shown in a state where the third sensors **101C** detect that the locking pins **120** have been fully extended into their locked position (e.g., as shown in FIG. **10**). At this point, the lift arm attachment safety system **100** may transmit one or more control signals via the vehicle drive control circuitry **106** to the drive control system **107** instructing the drive control system **107** to, for example, release the parking brake of the vehicle **105** thereby allowing the operator to drive the vehicle **105** and operate the lift arms **109** knowing that the adapter plate **108** is fully seated in the adapter frame **111** and the locking pins **120** are engaged.

[0040] In FIG. **17**, the lift arm attachment safety system **100** is shown in a state where an operator has provided a user input via the interface device **124** to request disengagement of the locking pins **120** so as to release the lift arm attachment **110**. Following receipt of the user input via the interface device **124**, the lift arm attachment safety system **100** may, again, transmit one or more control signals via the vehicle drive control circuitry **106** to the drive control system **107** instructing the

drive control system **107** to, for example, engage the parking brake of the vehicle **105**. Further, the lift arm attachment safety system **100** may transmit one or more control signals via the locking mechanism control circuitry **103** to the locking mechanism **104** instructing the locking mechanism **104** to retract the locking pins **120** (e.g., as shown in FIG. **9**) until the third sensors **101C** determine that the locking pins **120** have been fully retracted into their unlocked positions.

[0041] In FIG. **18**, the lift arm attachment safety system **100** is shown in a state where, the locking pins **120** have been retracted and the adapter plate **108** has been disengaged from the adapter frame **111** of the lift arm attachment **110** (e.g., as shown in FIG. **6**) such that the first sensors **101A** and second sensors **101B** no longer indicate that the adapter plate **108** is in proximity to the adapter frame **111**. At this point, the lift arm attachment safety system **100** may transmit one or more control signals via the vehicle drive control circuitry **106** to the drive control system **107** instructing the drive control system **107** to, for example, disengage the parking brake of the vehicle **105**.

[0042] One skilled in the art will recognize that the herein described components (e.g., operations), devices, objects, and the discussion accompanying them are used as examples for the sake of conceptual clarity and that various configuration modifications are contemplated. Consequently, as used herein, the specific exemplars set forth and the accompanying discussion are intended to be representative of their more general classes. In general, use of any specific exemplar is intended to be representative of its class, and the non-inclusion of specific components (e.g., operations), devices, and objects should not be taken as limiting.

[0043] The previous description is presented to enable one of ordinary skill in the art to make and use the invention as provided in the context of a particular application and its requirements. As used herein, directional terms such as “top,” “bottom,” “over,” “under,” “upper,” “upward,” “lower,” “down,” and “downward” are intended to provide relative positions for purposes of description, and are not intended to designate an absolute frame of reference. Various modifications to the described embodiments will be apparent to those with skill in the art, and the general principles defined herein may be applied to other embodiments. Therefore, the present invention is not intended to be limited to the particular embodiments shown and described, but is to be accorded the widest scope consistent with the principles and novel features herein disclosed.

[0044] With respect to the use of substantially any plural and/or singular terms herein, those having skill in the art can translate from the plural to the singular and/or from the singular to the plural as is appropriate to the context and/or application. The various singular/plural permutations are not expressly set forth herein for sake of clarity.

[0045] It is further contemplated that each of the embodiments of the method described above may include any other step(s) of any other method(s) described herein. In addition, each of the embodiments of the method described above may be performed by any of the systems described herein.

[0046] Those having skill in the art will recognize that the state of the art has progressed to the point where there is little distinction left between hardware and software implementations of aspects of systems; the use of hardware or software is generally (but not always, in that in certain contexts the choice between hardware and software can become significant) a design choice representing cost vs. efficiency tradeoffs. Those having skill in the art will appreciate that there are various vehicles by which processes and/or systems and/or other technologies described herein can be implemented (e.g., via hardware, software, and/or firmware), and that the preferred vehicle will vary with the context in which the processes and/or systems and/or other technologies are deployed. For example, if an implementer determines that speed and accuracy are paramount, the implementer may opt for a mainly hardware and/or firmware vehicle; alternatively, if flexibility is paramount, the implementer may opt for a mainly software implementation; or, yet again alternatively, the implementer may opt for some combination of hardware, software, and/or firmware. Hence, there are several possible vehicles by which the processes and/or devices and/or other technologies described herein may be implemented, none of which is inherently superior to

the other in that any vehicle to be utilized is a choice dependent upon the context in which the vehicle will be deployed and the specific concerns (e.g., speed, flexibility, or predictability) of the implementer, any of which may vary. Those skilled in the art will recognize that optical aspects of implementations will typically employ optically-oriented hardware, software, and/or firmware.

[0047] The foregoing detailed description has set forth various embodiments of the devices and/or processes via the use of block diagrams, flowcharts, and/or examples. Insofar as such block diagrams, flowcharts, and/or examples contain one or more functions and/or operations, it will be understood by those within the art that each function and/or operation within such block diagrams, flowcharts, or examples can be implemented, individually and/or collectively, by a wide range of hardware, software, firmware, or virtually any combination thereof. In one embodiment, several portions of the subject matter described herein may be implemented via Application Specific Integrated Circuits (ASICs), Field Programmable Gate Arrays (FPGAs), digital signal processors (DSPs), or other integrated formats. However, those skilled in the art will recognize that some aspects of the embodiments disclosed herein, in whole or in part, can be equivalently implemented in integrated circuits, as one or more computer programs running on one or more computers (e.g., as one or more programs running on one or more computer systems), as one or more programs running on one or more processors (e.g., as one or more programs running on one or more microprocessors), as firmware, or as virtually any combination thereof, and that designing the circuitry and/or writing the code for the software and/or firmware would be well within the skill of one of skill in the art in light of this disclosure. In addition, those skilled in the art will appreciate that the mechanisms of the subject matter described herein are capable of being distributed as a program product in a variety of forms, and that an illustrative embodiment of the subject matter described herein applies regardless of the particular type of signal bearing medium used to actually carry out the distribution. Examples of a signal bearing medium include, but are not limited to, the following: a recordable type medium such as a hard disk drive, a computer memory, solid state memory, cloud-based storage, and the like; and/or a transmission type medium such as a digital and/or an analog communication medium (e.g., a fiber optic cable, a waveguide, a wired communications link, a wireless communication link, etc.).

[0048] In a general sense, those skilled in the art will recognize that the various aspects described herein which can be implemented, individually and/or collectively, by a wide range of hardware, software, firmware, or any combination thereof can be viewed as being composed of various types of “electrical circuitry.” Consequently, as used herein “electrical circuitry” includes, but is not limited to, electrical circuitry having at least one discrete electrical circuit, electrical circuitry having at least one integrated circuit, electrical circuitry having at least one application specific integrated circuit, electrical circuitry forming a general purpose computing device configured by a computer program (e.g., a general purpose computer configured by a computer program which at least partially carries out processes and/or devices described herein, or a microprocessor configured by a computer program which at least partially carries out processes and/or devices described herein), electrical circuitry forming a memory device (e.g., forms of random access memory), and/or electrical circuitry forming a communications device (e.g., a modem, communications switch, or optical-electrical equipment). Those having skill in the art will recognize that the subject matter described herein may be implemented in an analog or digital fashion or some combination thereof.

[0049] Those skilled in the art will recognize that it is common within the art to describe devices and/or processes in the fashion set forth herein, and thereafter use engineering practices to integrate such described devices and/or processes into data processing systems. That is, at least a portion of the devices and/or processes described herein can be integrated into a data processing system via a reasonable amount of experimentation. Those having skill in the art will recognize that a typical data processing system generally includes one or more of a system unit housing, a video display device, a memory such as volatile and non-volatile memory, processors such as microprocessors

and digital signal processors, computational entities such as operating systems, drivers, graphical user interfaces, and applications programs, one or more interaction devices, such as a touch pad or screen, and/or control systems including feedback loops and control motors (e.g., feedback for sensing position and/or velocity; control motors for moving and/or adjusting components and/or quantities). A typical data processing system may be implemented utilizing any suitable commercially available components, such as those typically found in data computing/communication and/or network computing/communication systems. The herein described subject matter sometimes illustrates different components contained within, or connected with, other components. It is to be understood that such depicted architectures are merely exemplary, and that in fact many other architectures can be implemented which achieve the same functionality. In a conceptual sense, any arrangement of components to achieve the same functionality is effectively “associated” such that the desired functionality is achieved. Hence, any two components herein combined to achieve a particular functionality can be seen as “associated with” each other such that the desired functionality is achieved, irrespective of architectures or intermedial components. Likewise, any two components so associated can also be viewed as being “connected,” or “coupled,” to each other to achieve the desired functionality, and any two components capable of being so associated can also be viewed as being “couplable,” to each other to achieve the desired functionality. Specific examples of couplable include but are not limited to physically mateable and/or physically interacting components and/or wirelessly interactable and/or wirelessly interacting components and/or logically interacting and/or logically interactable components.

[0050] It will be understood by those within the art that, in general, terms used herein, and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes but is not limited to,” and the like). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to inventions containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should typically be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations. In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should typically be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, typically means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, and the like” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, and the like). In those instances where a convention analogous to “at least one of A, B, or C, and the like” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, and the like). It will be further understood by those within the art that virtually any disjunctive word and/or phrase presenting two

or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For example, the phrase “A or B” will be understood to include the possibilities of “A” or “B” or “A and B.”

[0051] It is believed that the present disclosure and many of its attendant advantages will be understood by the foregoing description, and it will be apparent that various changes may be made in the form, construction, and arrangement of the components without departing from the disclosed subject matter or without sacrificing all of its material advantages. The form described is merely explanatory, and it is the intention of the following claims to encompass and include such changes. [0052] Furthermore, it is to be understood that the invention is defined by the appended claims.

Claims

1. A lift arm attachment safety system comprising: an adapter plate receivable within an adapter frame of a lift arm attachment; one or more sensors disposed on the adapter plate and configured to detect a proximity of the adapter plate to the adapter frame; a drive control system configured to enable or disable the drive capabilities of a vehicle; at least one processor; and at least one memory device storing one or more instructions executable by the at least one processor for: receiving at least one signal from the one or more sensors indicating that the adapter plate is within a threshold proximity to the adapter frame; transmitting one or more control signals to the drive control system to cause the drive control system to disable drive capabilities of the vehicle in response to receiving the at least one signal from the one or more sensors indicating that the adapter plate is within a threshold proximity to the adapter frame.
2. The lift arm attachment safety system of claim 1, further comprising: a locking mechanism configured to control one or more locking pins; wherein the at least one memory device storing one or more instructions executable by the at least one processor further includes instructions for: receiving at least one signal from the locking mechanism indicating that the one or more locking pins are in a locked position; transmitting one or more control signals to the drive control system to cause the drive control system to enable drive capabilities of the vehicle in response to receiving the at least one signal from the one or more sensors indicating that the one or more locking pins are in a locked position.
3. The lift arm attachment safety system of claim 2, further comprising: an interface device configured to receive at least one control input from a user; wherein the at least one memory device storing one or more instructions executable by the at least one processor further includes instructions for: receiving at least one signal from the interface device indicating a control input from a user; transmitting one or more control signals to the drive control system to cause the drive control system to disable drive capabilities of the vehicle in response to receiving the at least one signal from the interface device indicating the control input from the user.
4. The lift arm attachment safety system of claim 3, wherein the at least one memory device storing one or more instructions executable by the at least one processor further includes instructions for: transmitting one or more control signals to the locking mechanism to cause the locking mechanism to move the one or more locking pins to an unlocked position in response to receiving the at least one signal from the interface device indicating the control input from the user.
5. The lift arm attachment safety system of claim 3, wherein the at least one memory device storing one or more instructions executable by the at least one processor further includes instructions for: receiving at least one signal from the one or more sensors indicating that the adapter plate is not within a threshold proximity to the adapter frame; transmitting one or more control signals to the drive control system to cause the drive control system to enable drive capabilities of the vehicle in response to receiving the at least one signal from the one or more sensors indicating that the adapter plate is not within a threshold proximity to the adapter frame.

- 6.** The lift arm attachment safety system of claim 1, further comprising: a locking mechanism configured to control one or more locking pins; wherein the at least one memory device storing one or more instructions executable by the at least one processor further includes instructions for: receiving the at least one signal from the one or more sensors indicating that the adapter plate is within the threshold proximity to the adapter frame; transmitting one or more control signals to the locking mechanism to cause the locking mechanism to move the one or more locking pins to a locked position in response to receiving the at least one signal from the one or more sensors indicating that the adapter plate is within the threshold proximity to the adapter frame.
- 7.** The lift arm attachment safety system of claim 1, wherein the drive control system includes at least one of a parking brake or a transmission.
- 8.** The lift arm attachment safety system of claim 1, wherein the one or more sensors disposed on the adapter plate and configured to detect a proximity of the adapter plate to the adapter frame includes: one or more first sensors configured to detect a proximity of at least one projection portion the adapter plate to a flange portion of the adapter frame; and one or more second sensors configured to detect a proximity of the adapter plate to a back plate of the adapter frame.
- 9.** The lift arm attachment safety system of claim 8, wherein the adapter plate includes: one or more apertures in the adapter plate configured to allow the one or more second sensors to detect the proximity of the adapter plate to the back plate of the adapter frame through the adapter plate.
- 10.** A lift arm attachment safety system comprising: at least one processor; and at least one memory device storing one or more instructions executable by the at least one processor for: receiving at least one signal from one or more sensors indicating that an adapter plate is within a threshold proximity to an adapter frame; transmitting one or more control signals to a drive control system to cause the drive control system to disable drive capabilities of a vehicle in response to receiving the at least one signal from the one or more sensors indicating that the adapter plate is within the threshold proximity to the adapter frame.
- 11.** The lift arm attachment safety system of claim 10, further comprising: wherein the at least one memory device storing one or more instructions executable by the at least one processor further includes instructions for: receiving at least one signal from a locking mechanism indicating that one or more locking pins are in a locked position; transmitting one or more control signals to the drive control system to cause the drive control system to enable drive capabilities of the vehicle in response to receiving the at least one signal from the one or more sensors indicating that the one or more locking pins are in a locked position.
- 12.** The lift arm attachment safety system of claim 11, further comprising: wherein the at least one memory device storing one or more instructions executable by the at least one processor further includes instructions for: receiving at least one signal from an interface device indicating a control input from a user; transmitting one or more control signals to the drive control system to cause the drive control system to disable drive capabilities of the vehicle in response to receiving the at least one signal from the interface device indicating the control input from the user.
- 13.** The lift arm attachment safety system of claim 12, wherein the at least one memory device storing one or more instructions executable by the at least one processor further includes instructions for: transmitting one or more control signals to a locking mechanism to cause the locking mechanism to move one or more locking pins to an unlocked position in response to receiving the at least one signal from the interface device indicating the control input from the user.
- 14.** The lift arm attachment safety system of claim 12, wherein the at least one memory device storing one or more instructions executable by the at least one processor further includes instructions for: receiving at least one signal from the one or more sensors indicating that the adapter plate is not within a threshold proximity to the adapter frame; transmitting one or more control signals to the drive control system to cause the drive control system to enable drive capabilities of the vehicle in response to receiving the at least one signal from the one or more sensors indicating that the adapter plate is not within a threshold proximity to the adapter frame.

15. The lift arm attachment safety system of claim 10, further comprising: wherein the at least one memory device storing one or more instructions executable by the at least one processor further includes instructions for: receiving the at least one signal from the one or more sensors indicating that the adapter plate is within the threshold proximity to the adapter frame; transmitting one or more control signals to a locking mechanism to cause the locking mechanism to move one or more locking pins to a locked position in response to receiving the at least one signal from the one or more sensors indicating that the adapter plate is within the threshold proximity to the adapter frame.

16. The lift arm attachment safety system of claim 10, wherein the transmitting one or more control signals to a drive control system to cause the drive control system to disable drive capabilities of a vehicle in response to receiving the at least one signal from the one or more sensors indicating that the adapter plate is within the threshold proximity to the adapter frame includes: transmitting one or more control signals to at least one of a parking brake or a transmission of the vehicle to cause the at least one of a parking brake or a transmission to disable drive capabilities of the vehicle in response to receiving the at least one signal from the one or more sensors indicating that the adapter plate is within the threshold proximity to the adapter frame.

17. A method for secure operation of a lift-arm vehicle comprising: receiving at least one signal from one or more sensors indicating that an adapter plate coupled to a lift-arm is within a threshold proximity to an adapter frame of a lift arm attachment; transmitting one or more control signals to a drive control system to cause the drive control system to disable drive capabilities of a vehicle in response to receiving the at least one signal from the one or more sensors indicating that the adapter plate is within the threshold proximity to the adapter frame.

18. The method of claim 17, further comprising: receiving at least one signal from a locking mechanism indicating that one or more locking pins are in a locked position; transmitting one or more control signals to the drive control system to cause the drive control system to enable drive capabilities of the vehicle in response to receiving the at least one signal from the one or more sensors indicating that the one or more locking pins are in a locked position.

19. The method of claim 18, further comprising: receiving at least one signal from an interface device indicating a control input from a user; transmitting one or more control signals to the drive control system to cause the drive control system to disable drive capabilities of the vehicle in response to receiving the at least one signal from the interface device indicating the control input from the user.

20. The method of claim 19, further comprising: transmitting one or more control signals to a locking mechanism to cause the locking mechanism to move one or more locking pins to an unlocked position in response to receiving the at least one signal from the interface device indicating the control input from the user.

21. The method of claim 19, further comprising: receiving at least one signal from the one or more sensors indicating that the adapter plate is not within a threshold proximity to the adapter frame; transmitting one or more control signals to the drive control system to cause the drive control system to enable drive capabilities of the vehicle in response to receiving the at least one signal from the one or more sensors indicating that the adapter plate is not within the threshold proximity to the adapter frame.

22. The method of claim 17, further comprising: receiving the at least one signal from the one or more sensors indicating that the adapter plate is within the threshold proximity to the adapter frame; transmitting one or more control signals to a locking mechanism to cause the locking mechanism to move one or more locking pins to a locked position in response to receiving the at least one signal from the one or more sensors indicating that the adapter plate is within the threshold proximity to the adapter frame.
